

TaskFlow — Production-Grade Flutter Technical Assessment Application

TaskFlow is a project management mobile application built with **Flutter & Dart** following **Clean Architecture (SOLID principles)**, reactive state management using **Riverpod**, simulated authentication with JWT access/refresh token rotation, local storage caching, role-based access control (RBAC), and interactive evaluator debugging controls.

Complete End-to-End User Flow & Screen Navigation

TaskFlow guides users through a clear, intuitive project management workflow:

```
graph TD
    Splash[1. Splash Screen - Session Check] -->|Session Restored| Dashboard[3. Dashboard / Home Screen]
    Splash -->|No Valid Token| Login[2A. Login Screen]
    Login -->|Tap Register| Register[2B. Register Screen]
    Register -->|Account Created| Dashboard
    Login -->|Authenticated| Dashboard

    Dashboard -->|Tap View All Projects| ProjectsList[4. Projects List Screen]
    Dashboard -->|Tap Project Card| ProjectDetail[5. Project Detail Screen]
    Dashboard -->|Tap Open Task Board| TaskBoard[6. Task List Board & Multi-Filter]
    Dashboard -->|Tap Notifications Icon| Notifications[10. Notifications Inbox]
    Dashboard -->|Tap Profile/Settings Icon| ProfileSettings[9. Profile & Evaluator Debug Settings]

    ProjectsList -->|Tap Project| ProjectDetail
    ProjectsList -->|Admin: Tap + New Project| CreateProjectModal[Create Project Modal]

    ProjectDetail -->|Tap Task Card| TaskDetail[7. Task Detail Screen]
    ProjectDetail -->|Tap + Add Task| CreateTask[8. Create / Edit Task Screen]

    TaskBoard -->|Tap Task Card| TaskDetail
    TaskBoard -->|Tap + New Task| CreateTask
    TaskBoard -->|Tap Filter Icon| FilterModal[Multi-Filter Bottom Sheet]

    TaskDetail -->|Tap Edit Icon| CreateTask
    TaskDetail -->|Tap Assignee Card| AssigneeModal[Org Member Selector Modal]
    TaskDetail -->|Type comment| CommentThread[Comments & Activity Log]

    ProfileSettings -->|Toggle Dark/Light Switch| DynamicTheme[Instant Theme Switch]
    ProfileSettings -->|Toggle Debug Switches| ErrorSimulation[Simulated 404 / 408 / 422 / Offline Mode]
    ProfileSettings -->|Tap Manual Token Refresh| JWTTokenRefresh[Simulated JWT Expiry & Refresh]
    ProfileSettings -->|Tap Sign Out| Login
```

Screen-by-Screen User Guide

1. Splash Screen (splash_screen.dart)

- **How it works:** Automatically checks secure storage (SecureTokenStorage) on application startup.
- **Flow:** If a valid session or refreshable token exists, auto-logs the user in and navigates to the **Dashboard Screen**. If no session exists, navigates to the **Login Screen**.

2. Login & Registration Screens (login_screen.dart & register_screen.dart)

- **Client-Side Validation:** Validates required inputs, email format, and password length.
- **Password Eye Toggle:** Eye icon button (visibility_off / visibility) on password fields allows users to toggle password visibility (hidden by default).
- **Evaluator Quick-Fill Chips:** One-tap test credential chips on the Login screen to log in instantly as Org A Admin, Org A Member, Org B Admin, or Org B Member.

3. Dashboard / Home Screen (dashboard_screen.dart)

- **Organization Header:** Displays active user name, email, avatar, and Role badge (Admin vs Member).
- **Task Summary Breakdown:** Live counters for tasks in To Do, In Progress, In Review, and Completed.

- **Top App Bar Quick Actions:**

- 🌙 / ☀️ **Theme Toggle:** Instantly switch between Light Mode and Dark Mode.
- 🔔 **Notifications Icon:** Badge counter showing unread assignment notifications.
- ⚙️ **Settings Icon:** Opens Profile & Evaluator Debug Settings.

4. Projects List Screen (`project_list_screen.dart`)

- **Org-Scoped Projects:** Displays projects belonging to the logged-in user's organization (`org_id`).
- **Role-Based Admin Guard:** Only Organization Admins (`org_admin`) see the + New Project floating action button and project delete actions.

5. Project Detail Screen (`project_detail_screen.dart`)

- **Overview Card:** Displays project description and status badge.
- **Mini Task Summary Grid:** Live counts of project tasks grouped by status.
- **Project Tasks:** Filtered list of tasks specific to this project.

6. Task List Board & Multi-Filter (`task_list_screen.dart`)

- **Live Search Bar:** Search tasks by title or keyword.
- **Multi-Criteria Filter Sheet:** Filter tasks by:
 - **Status** (`todo`, `in_progress`, `review`, `done`)
 - **Priority** (`low`, `medium`, `high`, `urgent`)
 - **Assignee** (filtered to active organization members)
 - **Due Date Range** (Interactive Date Range Picker)
- **Active Filter Chips:** Horizontally scrollable chips showing active filters with one-tap removal.

7. Task Detail Screen (`task_detail_screen.dart`)

- **Inline Pickers:** Tap status badge or priority badge to change task status/priority inline.
- **Assignee Selector Modal:** Tap Assignee card to assign or unassign team members.
 - *Business Logic Guard:* Prevents assigning members from outside the active organization.
- **Comments & Activity Thread:** View comments and post new comments in real-time.

8. Create / Edit Task Screen (`create_edit_task_screen.dart`)

- Form to create new tasks or update existing tasks (Title, Description, Project, Priority, Status, Assignee, and Due Date Picker).

9. Profile & Evaluator Debug Settings (`profile_settings_screen.dart`)

- **User Profile Card:** Displays user avatar, email, organization ID, and role.
- **Appearance Settings:** Toggle between Light Theme and Dark Theme.
- **Simulated JWT Session Status:** Displays token expiry time, auto-refresh status, and a button to trigger manual token refresh.
- **Evaluator Debug Control Panel:** Interactive toggles for:
 - **Simulate Offline Mode** (Shows sticky warning banner and loads cached data from `SharedPreferences`).
 - **Simulate 404 Not Found**
 - **Simulate 408 Network Timeout**
 - **Simulate 422 Payload Validation Error**
 - **Artificial Network Latency Slider** (0 – 1500 ms)

10. Notifications Inbox (`notifications_screen.dart`)

- List of task assignment notification alerts. Tapping a notification marks it as read and opens the corresponding task.

🔑 Test Credentials Table

Reviewers can test role-based permissions and multi-organization behavior using these credentials:

Organization	Role	Name	Email	Password
Nimbus Digital (<code>org_a1b2c3</code>)	Admin	Aditya Sharma	<code>aditya.admin@nimbusdigital.test</code>	Password123!
Nimbus Digital (<code>org_a1b2c3</code>)	Member	Ramesh Kumar	<code>ramesh.member@nimbusdigital.test</code>	Password123!

Nimbus Digital (org_a1b2c3) Member Rinki Patel rinki.member@nimbusdigital.test Password123!
Harborlight Studios (org_d4e5f6) Admin Shivani Verma shivani.admin@harborlightstudios.test Password123!
Harborlight Studios (org_d4e5f6) Member Arzzoo Khan arzzoo.member@harborlightstudios.test Password123!

💡 Quick Fill: Tap any credential chip on the Login screen to auto-fill email and password!

🏗 Architecture & Layering Overview

TaskFlow strictly enforces **Clean Layered Architecture**:

```
d:\Assignment\  
├── assets/  
│   └── mock_data/  
│       └── TaskFlow-MockData.json  
├── lib/  
│   ├── core/  
│   │   ├── constants/      # AppColors, AppTypography  
│   │   ├── errors/        # Sealed Failures hierarchy  
│   │   ├── theme/         # Light & Dark Material 3 ThemeData  
│   │   └── utils/          # Functional Result<T> pattern  
│   ├── data/  
│   │   ├── datasources/   # MockDataSource, LocalStorageDataSource, SecureTokenStorage  
│   │   ├── models/        # DTOs with JSON serialization  
│   │   └── repositories/   # Repository implementations  
│   ├── domain/  
│   │   ├── entities/      # Pure Dart Domain Entities (TaskEntity, Project, User)  
│   │   └── repositories/   # Abstract Repository Interfaces  
│   ├── presentation/  
│   │   ├── providers/     # Riverpod Providers (Auth, Projects, Tasks, Theme, Debug)  
│   │   ├── screens/       # All 11 application screens  
│   │   └── widgets/       # Reusable Badges, Buttons, Inputs, Dialogs  
│   └── main.dart  
├── test/  
│   ├── unit/              # Repository, Auth, and Filter unit tests  
│   ├── widget/            # Login form & badge widget tests  
│   └── integration/       # App launch & integration tests
```

🔧 How to Run & Test

Prerequisites

- Flutter SDK: ^3.32.5
- Dart SDK: ^3.8.1

Commands

1. Install Dependencies:

```
flutter pub get
```

2. Run Application:

```
flutter run
```

3. Run Automated Test Suite:

```
flutter test
```

4. Build Production Release APK:

```
flutter build apk --release
```

🔧 How to Test Evaluator Debug & Error Features

1. Log in with any test credential (e.g. aditya.admin@nimbusdigital.test).

2. Open **Profile & Settings** (gear icon in Dashboard top right).
3. Under **Evaluator Debug Control Panel**:
 - **Offline Mode**: Toggle on "Simulate Offline Mode". Notice the sticky warning banner at the top of screens. Data continues loading smoothly from local `SharedPreferences` cache.
 - **404 / 408 / 422 Errors**: Toggle any error switch, then return to Dashboard or Tasks and pull-to-refresh to view the retryable error UI state.
 - **Manual Token Refresh**: Tap "Simulate Manual Token Refresh Now" to issue a new JWT access token.
 - **Dark Mode**: Tap the Theme toggle switch to switch between Light Mode and Dark Mode.